Anodyne

0.600 / MATH SPECIMEN

Mathematics, from a line of text to a page of notation.

The complete Stack Exchange / OverLyX test expressions, with Anodyne's roots, delimiters, mathematical alphabets and named stylistic choices.

01 Complex analysis

Residue-theorem specimen. Suppose f is holomorphic throughout G apart from finitely many isolated singularities $a_1, a_2, \dots, a_m$. A rectifiable closed path γ avoids these points and is null-homologous, $\gamma \approx 0$, in G. The test deliberately nests a bold letter and a double-struck space inside the exponent:

$$\frac{1}{2\pi i}\int_{\gamma} f\left(x^{\mathbf{N}\in\mathbb{C}^{N\times 10}}\right) = \sum_{k=1}^{m} n(\gamma; a_k)\operatorname{Res}(f; a_k)\,.$$

Maximum modulus. On a bounded domain $G \subset \mathbb{C}$, let f extend continuously to the closure G^{-} and be holomorphic inside. Its maximum occurs on the boundary:

$$\max\{\,|f(z)| : z \in G^{-}\,\} = \max\{\,|f(z)| : z \in \partial G\,\}\,.$$

02 Large operators, in text and on display

At text size, $\iiint_{Q} f(x,y,z)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$ and $\prod_{\gamma\in\Gamma_c} \partial(\tilde{X}_\gamma)$ stay part of the sentence. A display brings out the larger drawings:

$$\iiiint_{Q} f(w,x,y,z)\,\mathrm{d}w\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \le \oint_{\partial Q} f'\left(\max\left\{\frac{\|w\|}{|w^2+x^2|};\quad \frac{\|z\|}{|y^2+z^2|};\quad \frac{\|w\oplus z\|}{|x\oplus y|}\right\}\right).$$

03 Accents and mathematical alphabets

$$\hat{a}\quad \tilde{b}\quad \bar{c}\quad \vec{v}\quad \dot{x}\quad \ddot{y}\quad \breve{u}\quad \check{z}\quad \acute{e}\quad \grave{e}\qquad \hat{A}\quad \tilde{N}\quad \bar{X}\quad \dot{\Phi}$$

$$\widehat{xyz}\quad \widetilde{abc}\quad \overline{z+w}\qquad a \neq b\quad x \not< y\quad \varepsilon \neq \epsilon\quad \varphi \neq \phi$$

$$\mathscr{ABCL}\qquad \mathbb{RNC}\qquad \mathfrak{g}\qquad \boldsymbol{\alpha}\cdot\mathbf{v}\qquad \mathbf{ABCxyz}\qquad \mathtt{01Il}$$

Greek roundhand, with accents attached by the font's MATH data:

$$\hat{\iota}\quad \ddot{\iota}\qquad \hat{\lambda}\quad \ddot{\lambda}\qquad \hat{\mathit{\lambda}}\quad \ddot{\mathit{\lambda}}\qquad \mathit{\alpha}+\mathit{\Gamma}+\mathit{\varepsilon}+\mathit{\phi}$$

04 Fractions, roots and growing delimiters

$$\sum_{k=1}^{n} \frac{\alpha_k x^{2k}}{\sqrt{1+\beta_k^2}} \le \int_{-\infty}^{\infty} \left|\frac{\partial \Phi}{\partial t}\right| \mathrm{d}t$$

$$\left\{ \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \\ a_3 & b_3 \end{bmatrix} \prod_{i=1}^{m} \lambda_i \right\}^{1/m} \qquad \sqrt[3]{\hbar+\varepsilon} \qquad f_{i_j}^{a^b} + \overbrace{a+b+c}^{n}$$

05 Six choices, one consistent style

Default above; upright integrals, slanted inequalities, compact sets, vertical negations, oval empty set and lowered-bar $\hbar$ below.

$$\int_0^1 x^2\,\mathrm{d}x \qquad a \le b \qquad x \in A \qquad A \neq B \qquad A \cap B = \varnothing \qquad E = \hbar\omega$$

$$\int_0^1 x^2\,\mathrm{d}x \qquad a \leqslant b \qquad x \in A \qquad A \neq B \qquad A \cap B = \emptyset \qquad E = \hbar\omega$$

Mathematical test expressions adapted from Henri Menke, Which OpenType math fonts are available? (TeX–LaTeX Stack Exchange, 2018; CC BY-SA 3.0), as used by OverLyX. Explanatory wording and layout changed; Anodyne examples added. Source: https://tex.stackexchange.com/q/425098. Adapted specimen: https://creativecommons.org/licenses/by-sa/4.0/. Typeset with LuaLaTeX and unicode-math. Font software: SIL OFL 1.1.